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The Mathematics and Design of CobbleShoal

Technical Report

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Abstract

CobbleShoal is the second detector built for Strawberry Creek, and it exists because DuskCrayfish is structurally blind to a sensor that stops moving. Where DuskCrayfish treats a monitoring site as one object carrying a vector of readings, CobbleShoal treats each individual sensor as its own node in a nested graph, so a conductivity probe and the depth probe beside it are separate nodes that talk to each other through a learned variable graph and to the rest of the creek through a directed flow graph. On top of that shared trunk it runs four detection channels rather than one, each answering a different question about the same window, and it turns each channel into a right-tail probability against that channel's own fault-free distribution before combining them. No labelled anomalies exist for this creek and none will, so nothing here is tuned against a fault: the alerting threshold comes from Extreme Value Theory at a false alarm rate chosen in advance. This document sets out the mathematics of the encoder, the four channels, the calibration, and the reasoning behind each choice.

1 Introduction

DuskCrayfish learns what the creek normally looks like and flags what it cannot predict. That works well for an onset, a discharge arriving and pushing conductivity somewhere the model did not expect. It fails for a fault that stops the reading moving at all. If a probe freezes, the model's one-step forecast is anchored on the frozen reading, it predicts the frozen number, the reading is the frozen number, and the residual is zero. A frozen sensor is therefore invisible to a forecast residual, not approximately but exactly, and no threshold on that residual can recover it.

CobbleShoal is built around that gap. Everything below follows from two ideas. The first is that a sensor, not a site, is the right unit: the creek is a graph of individual probes, and a probe that has stopped reporting is a different object from the site it sits at. The second is that one number cannot answer every question about a window, so the model produces several, each anchored somewhere different, and combines them only after each has been made comparable.

1.1 The deployed configuration

The values in Table 1 are the settings the shipped checkpoint runs with, read from the model definition and the calibration artifact beside the weights. They are referred to throughout the mathematics that follows so the general equations stay tied to this specific model.

Table 1: The deployed configuration. These are the settings the current checkpoint and its calibration artifact carry, not defaults or recommendations.

Setting	Value
Sites S	5
Canonical variables V	5
Sensor nodes N	17
Window T	24 timesteps (6 hours at 15 min sampling)
Context features	2 (hour sine and cosine)
Hidden dimension d	64
Variable embedding	8
Site and node embedding	12
Transport hops K	2
Leave-one-out weight λ_{loo}	0.5
Reconstruction weight λ_{rec}	0.5
Node dropout p	0.15
Scale floor	10^{-2}
Anchor staleness limit	4 timesteps
Dispersion window	8 timesteps
Trainable parameters	63,605
Null tail percentile	95
POT initial threshold percentile	98
Operating false alarm rate q	10^{-4}
Calibrated threshold z_q	39.130

Sixty-three thousand parameters is small by the standards of the architectures this design borrows from, and it is small for the same reason DuskCrayfish is small. The corpus is a rolling window of one creek's recent behavior, not an archive, and a larger model on that much data memorizes the recent past rather than learning the creek's rhythm. The parameter count is dominated by the hidden dimension of 64, which is what the sibling and transport graphs need in order to carry a useful message, rather than by depth.

2 Data and Notation

2.1 The nested node graph

DuskCrayfish's graph has one node per monitoring site, carrying a vector of that site's readings. CobbleShoal splits that apart. A node is a (site, variable) pair, so the conductivity probe at north_fork_0 and the temperature probe at north_fork_0 are two different nodes. The site inventory says which pairs exist, and it is uneven by design: footbridge carries five variables because it has a dissolved oxygen and floating conductivity sensor the other four sites do not, so the deployed graph has $N = 17$ nodes over $S = 5$ sites rather than $S \times V = 25$. Nodes that do not exist are not modelled as missing; they are absent from the registry.

This is the structural choice the rest of the design rests on, and it is worth being explicit about what it buys. A frozen conductivity probe is now a node whose own readings have stopped while its siblings at the same site, and its neighbours downstream, keep moving. That contrast is measurable. In a site-level graph the frozen probe is one component of one node's feature vector, averaged in with the readings that are still fine, and the contrast has nowhere to appear. The nested formulation follows the same idea as graph architectures that propagate over variables and locations separately [Felice et al., 2024].

Two graphs connect the nodes. The variable graph A_{var} joins nodes that sit at the same site, so a probe can hear its own siblings. The flow graph joins sites in the direction water actually moves, and is described in Section 3.3.

2.2 Staleness, and what is deliberately not an input

Readings arrive on a 15 minute grid, but loggers do not respect it. A reading within half a grid step of a grid point is that grid point's target, which recovered six times footbridge's usable supervision, because its

Table 2: Notation.

Symbol	Meaning	Dimension
S	monitoring sites	scalar, $S = 5$
V	canonical variables	scalar, $V = 5$
N	sensor nodes, one per (site, variable) that exists	scalar, $N = 17$
T	timesteps in the input window	scalar, $T = 24$
d	hidden dimension	scalar, $d = 64$
K	transport hops	scalar, $K = 2$
x_t	node readings at step t	N
δ_t	staleness, steps since this node last reported	N
$\boldsymbol{h}_t$	node hidden state	$N \times d$
A_{var}	same-site adjacency over nodes	$N \times N$
W	junction share matrix over sites	$S \times S$
P	receive matrix, $W^\top$	$S \times S$
μ, σ	predicted mean and scale	N
c_j	channel j's score	N
p_j	channel j's right-tail probability	N
Λ	combined Fisher statistic	N
q	false alarm rate chosen in advance	scalar, 10^{-4}
z_q	alerting threshold at rate q	scalar, 39.130

logger reports at :17 and :28 rather than :15. Between reports the last value is carried forward, and δ_t counts how many steps have passed since the node actually reported. The model reads $\log(1 + \delta_t)$ as a feature.

The binary observation mask is deliberately not an input. Feeding the mask in lets a model exploit informative missingness, which is the right move where a missing measurement is itself a decision, as in clinical records [Che et al., 2018]. Here missingness is flat batteries and network outages. A model given the mask learns that offline implies anomalous, and then every power failure in the catchment arrives as a discharge alert. The elapsed time is a feature; the fact of absence is not.

3 Model Architecture

All of the scoring heads sit on one shared encoder. Per timestep, a node's value is decayed by its variable's learned half life and masked if the node is hidden, concatenated with log staleness, the clock context and the variable, site and node embeddings, projected to d dimensions, multiplied by a static gate built from the site inventory and the site embedding, passed through the decay LSTM, normalized, and exchanged with its siblings. Once at the end of the window, node states are pooled to sites, transported downstream, and mixed back into the nodes.

3.1 The decay LSTM

Recurrent cells assume evenly spaced observations. These are not evenly spaced, so both the incoming value and the carried state decay with elapsed time [Hochreiter and Schmidhuber, 1997]. For a node of variable v with staleness δ,

$$\gamma_v^x(\delta) = 2^{-\delta/\tau_v^x}, \qquad \gamma^h(\delta) = 2^{-\delta/\tau^h}, \tag{1}$$

where τ_v^x is a learned half life per variable and τ^h a learned half life per hidden unit. The state update is the ordinary LSTM cell applied to the decayed state,

$$(\boldsymbol{h}_t, \boldsymbol{c}_t) = \text{LSTMCell}\big(\boldsymbol{u}_t,\ (\boldsymbol{h}_{t-1} \odot \gamma^h,\ \boldsymbol{c}_{t-1})\big). \tag{2}$$

The parameterisation matters more than it looks. Both half lives are stored as $\log \tau$ and exponentiated. An earlier form used $\exp(-\delta \cdot |w|)$ initialised at $w = 0$, where the absolute value has no gradient at zero, so nothing ever moved w off it and the decay was not merely inert but dead. A log half life is positive everywhere, differentiable everywhere, and initialises to a stated prior: four hours for conductivity, one hour for stage, three for temperature. Those priors are starting points the loss is free to move, not claims about the creek.

3.2 The sibling exchange

After the recurrent step, each node hears the other probes at its own site. For node states $\boldsymbol{h}$ and a liveness vector ℓ,

$$\boldsymbol{m}_i = \frac{\sum_j (A_{\text{var}})_{ij}\, \ell_j\, \boldsymbol{W}_s \boldsymbol{h}_j}{\max\left(\sum_j (A_{\text{var}})_{ij}\, \ell_j,\ \epsilon\right)}, \qquad \boldsymbol{h}'_i = \boldsymbol{W}_x\,[\,\boldsymbol{h}_i \,\|\, \boldsymbol{m}_i\,]. \tag{3}$$

The denominator is the count of live senders, and where a node has none the message is exactly zero rather than a small number, so a node whose whole site is offline is not fed an average of nothing.

Liveness gates who talks, not what is said. It is read from raw staleness, which is timing rather than measurement, and that distinction is what lets the leave-one-out channel in Section 5.1 be honest: hiding a node removes every path to its own readings while leaving intact the question of whether its neighbours are currently reporting.

The exchange runs at every timestep rather than once at the end. It costs about 1.4 times the epoch time, it does not improve mean error, and it was the more reliable configuration on every seed tried, which is the only reason it is the one that ships.

3.3 Directed downstream transport

Node states are pooled to their sites, and site states are moved downstream along the creek's flow edges. Transport is forward only. Writing W for the matrix whose entry W_{ab} is the share of site b's inflow contributed by site a, and $P = W^\top$ for the receive matrix, the transported context is

$$\boldsymbol{z} = \sum_{k=1}^{K} \theta_k\, \widetilde{P^k}\, \boldsymbol{W}_m \boldsymbol{g}, \tag{4}$$

where $\boldsymbol{g}$ are the pooled site states and $\widetilde{\cdot}$ denotes the same live-sender normalization used for siblings. This is the discrete analogue of tail-up weighting on a stream network, where influence moves with the water and splits at junctions [Ver Hoef and Peterson, 2010].

The junction split is learned, not assumed. Tail-up weighting properly wants a discharge ratio; nothing in this data measures discharge, and a hardcoded even split is an assumption the model then has to work around. A softmax over each junction's contributors makes the split something the data fits, and the fitted value becomes a physical claim a field survey can go and check.

The reverse path was removed. It had a real argument behind it, that an upstream site and its downstream neighbour share a storm and a baseflow regime, so a downstream reading is informative about an upstream one even though water does not flow that way. The measurement settled it: detection was bit-identical with the reverse weights zeroed, so the reverse path was carrying nothing the forward path and the sibling graph did not already carry.

Forward-only transport has a property worth checking on the trained weights. The forward matrix is nonzero only along downstream paths, so on a tree it never connects the north fork to the south fork at any number of hops, and cross-fork response is exactly zero rather than merely small.

3.4 Local effects

Each node, site and variable carries a learned embedding. These are what let one shared trunk represent five sites that behave differently without a separate model per site, which is the role trainable node embeddings play in graph-based spatiotemporal forecasting generally [Cini et al., 2023]. A masked node keeps its embeddings and its clock context, so it still knows which sensor at which site it is and what time it is; what it loses is every path to its own readings.

4 Training

Every head is trained under a Gaussian negative log likelihood rather than squared error. For a prediction μ, a predicted scale σ, a target y and a mask M marking the entries where the term can honestly be computed,

$$\mathcal{L}(\mu, \sigma, y) = \frac{1}{|M|} \sum_{i \in M} \left[\tfrac{1}{2} \left(\frac{y_i - \mu_i}{\sigma_i} \right)^2 + \log \sigma_i \right]. \tag{5}$$

The $\log \sigma$ term is what stops the model claiming certainty everywhere, and it is what makes the dispersion channel in Section 5.1 mean anything at all. The scale is floored at 10^{-2}, because without a floor the likelihood rewards driving σ toward zero on any node the model happens to fit well, which then blows up the dispersion channel that the stuck-sensor detector depends on.

The full objective has three terms,

$$\mathcal{L} = \mathcal{L}_{\text{forecast}} + \lambda_{\text{loo}}\,\mathcal{L}_{\text{loo}} + \lambda_{\text{rec}}\,\mathcal{L}_{\text{rec}}, \qquad \lambda_{\text{loo}} = \lambda_{\text{rec}} = 0.5. \tag{6}$$

Combining a forecasting objective with a reconstruction objective follows the same reasoning as multivariate anomaly detectors that train both heads [Zhao et al., 2020]: the two disagree in different ways, and a fault that hides from one need not hide from the other.

The forecast and leave-one-out terms are computed on two separate forward passes, and the separation is not cosmetic. The leave-one-out term hides each node with probability $p = 0.15$ and asks the model to predict it from its siblings, its flow neighbours and the clock. Applying that node dropout to a single shared pass moved nominal 68/95 interval coverage from 71/94 to 90/100, because the model hedged its forecast scale against inputs that might be masked. Two passes and two scale heads keep the forecast scale answering how wrong the next-step forecast is, and the leave-one-out scale answering how wrong the model is about a node it cannot see. Those are different questions with different answers, and one head cannot hold both.

The leave-one-out term is scored only where a node was actually hidden. Scoring the visible ones would be the forecast objective again under a second head.

5 Detection

5.1 Four channels, anchored differently

The model produces four numbers per node per window. What separates them is what each is anchored on, and that is the whole design.

Excursion is the standardised one-step forecast residual, $|y - \mu|/\sigma$, computed only where the target is observed and the anchor is fresh. Scoring a graph model's forecast residual is the usual construction for detecting a multivariate series that has left its normal relationship with its neighbours [Deng and Hooi, 2021]. It catches a reading that moved when it should not have. It is structurally blind to a frozen sensor, for the reason given in Section 1, and that blindness is why the other three exist.

Leave-one-out is $|y - \mu_{\text{loo}}|/\sigma_{\text{loo}}$, where the prediction is made with the node's own input removed. With the node hidden the prediction comes from its siblings, its flow neighbours and the clock, all of which are still moving, so a frozen reading stands out by exactly as much as the creek has moved since it froze. Anchor freshness is not a condition here, because the leave-one-out prediction never touches the node's anchor.

Reconstruction is the mean standardised error between the encoder's reconstruction of the input window and the window itself, over observed steps. It is anchored on neither the last value nor the neighbours, which makes it the only channel that can see a window whose shape is wrong while its level is right and the network agrees.

Dispersion asks how much flatter the sensor is than the model expected,

$$c_{\text{disp}} = \max\left(0,\ \log \frac{\max(\sigma, \phi)}{\max(\hat{s}, \phi)}\right), \tag{7}$$

where $\hat{s}$ is the observed standard deviation over the last 8 steps and $\phi = 10^{-3}$ is a floor. It costs nothing, it has no head of its own, and it is what a bit-exact freeze trips: the residual goes to zero, which is what blinds excursion, but the observed dispersion goes to zero while the predicted scale does not.

Dispersion requires real observations in its window, and that requirement was learned the hard way. Values are carried forward, so a node that is simply offline is flat for exactly the same arithmetic reason a frozen one is. Without the observation requirement, ordinary outages sat in this channel's fault-free null at the same magnitude as the faults, the null's tail swallowed them, and detection at the operating threshold on stuck faults measured 0.0% while the same channel was still ranking the frozen node first.

5.2 Nulls and p-values

Raw channel scores are not comparable. An excursion of 3 and a dispersion of 3 mean different things, and a maximum over raw magnitudes lets an uninformative channel outrank an informative one. This is not

a hypothetical: a measured run scored the maximum rule at 68.6% where a single channel alone reached 92.2%.

Each channel is therefore turned into a right-tail probability against its own fault-free distribution. Below the 95th percentile the empirical distribution is used directly,

$$p = \frac{1 + \#\{s_i \geq s\}}{1 + n}. \tag{8}$$

The +1 keeps a score above every null sample from reading as exactly zero, which would dominate any combination rule outright. Above the 95th percentile the tail is a fitted Generalised Pareto, so two scores that both sit past every observed sample still order against each other rather than both saturating at the $1/(n+1)$ floor. Fitting the excesses over a high threshold with a Generalised Pareto is the standard limit result for tails [Pickands III, 1975], and the fit uses a one-dimensional profile likelihood [Grimshaw, 1993].

The tail used for p-values clamps the shape parameter γ at zero, and this is deliberate. A negative γ is a finite upper endpoint, the maximum likelihood estimate puts that endpoint at the largest observed excess, and every score past it collapses to $p = 0$. That is not a small p-value, it is a tie: the largest fault-free sample and a sensor frozen for six hours both land on the floor, the combined score saturates for both, and no threshold can separate them. Clamping to the exponential limit keeps the tail unbounded, keeps the ordering the combination rule needs, and is conservative, since it assigns a larger p than the unconstrained fit. The unconstrained γ is still stored and reported.

5.3 Combination

The per-channel probabilities are combined into one statistic per node [Fisher, 1925],

$$\Lambda = -2 \sum_j \log p_j. \tag{9}$$

Fisher's method assumes the channels are independent, and these are not. Dispersion and leave-one-out both fire on a frozen sensor, so an agreeing pair overstates its significance. The consequence is bounded and worth stating precisely: the statistic is still monotone in the evidence, and the degrees of freedom are constant across nodes for a given model, so the ranking is unaffected. Only the absolute p-value is optimistic. That is exactly why the alerting threshold does not come from the chi-squared tail.

5.4 The threshold

The threshold is fitted to the combined statistic's own fault-free distribution by peaks over threshold [Siffer et al., 2017]. An initial threshold u sits at the 98th percentile of the fault-free statistics, a Generalised Pareto is fitted by maximum likelihood to the N_u exceedances above it, and

$$z_q = u + \frac{\sigma}{\gamma}\left[\left(\frac{q\,n}{N_u}\right)^{-\gamma} - 1\right], \qquad z_q = u - \sigma \log\left(\frac{q\,n}{N_u}\right) \quad \text{as } \gamma \to 0. \tag{10}$$

The deployed fit has $u = 19.860$, $N_u = 68$ exceedances out of $n = 3400$ fault-free windows, $\sigma = 3.637$, $\gamma = 0$, and therefore $z_q = 39.130$ at $q = 10^{-4}$.

The value of q is chosen in advance and the false alarm rate follows by construction, with no labels anywhere in the procedure. This is the property the whole calibration exists to preserve, and it is why the nulls and the threshold are fitted once at calibration time, written to an artifact beside the weights, and loaded unchanged at inference. A detector that refits its null on the window it is judging does not have a null; it has a description of the data it just saw.

5.5 What the alerting path can see

The detector is handed one number per node and nothing else. Channels are an implementation detail of how that number is built, and every caller that learned their names became a place the model could not change. Diagnostics that genuinely want the per-channel scores can ask for them explicitly, and nothing on the alerting path does.

Rain is handled outside this model entirely. The rain gate moves the alerting threshold and never the score, for the reason set out in that component's own documentation: the threshold was fitted against the fault-free

distribution of the combined statistic, and a statistic scaled by a rain factor is no longer a draw from the distribution that fit was made on. CobbleShoal declares no built-in support behaviour, and it declares it as an empty list rather than omitting the declaration, so the record says the question was asked.

6 Limitations and Future Work

The four channels are not independent and the combination rule assumes they are. The ranking survives this, as argued in Section 5.3, and the threshold is fitted to the statistic's own null rather than to a chi-squared tail precisely so the dependence does not reach the alerting decision. What is still unavailable is any statement about how much evidence a combined score represents, as opposed to how it ranks against other nodes.

The measured alarm rate at the operating q ran above nominal on the calibration corpus. The calibration artifact records this. Whether that is finite-sample noise in the tail fit, dependence between channels leaking into the combined null, or a genuine mismatch between the fault-free corpus and the verification set is open, and it is a question for the group rather than a defect in any one file.

The junction split is learned but has never been checked against a field survey. Its value is a physical claim about how much of oxford's inflow arrives from each fork, and it is currently unverified.

The site inventory is uneven, and the two channels that only footbridge carries are the two that would separate disturbance types. Dissolved oxygen and floating conductivity exist at exactly one of five sites, so any downstream reasoning that needs them can be attempted at one site and nowhere else.

Finally, every number in this document describes the shipped checkpoint and the calibration artifact fitted alongside it. The nulls and the threshold are fitted quantities, and they will move when the model is refitted.

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